Week 6 Lab – API Gateway Security Report
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Environment

- **Cloud Provider Used:** Cloudflare Workers & Cloudflare API Gateway
- **Backend API:** A simple posts API *(/api/v1/posts/allposts)* hosted as a Cloudflare Worker
- **Tools Used:** Cloudflare dashboard, VS Code, Postman, curl

Objective

The goal of this lab was to create an API Gateway to securely expose a backend API while enforcing authentication and basic security controls.

Implementation Details

1. Authentication

- **Method:** JWT-based authentication
- **Implementation:**
  - Incoming requests to the API Gateway must include a valid JWT in the Authorization header.
  - JWTs were verified for:
    - Signature validity using JWKS endpoint
    - Expiration (exp)
    - Issuer (iss) and audience (aud)
- Invalid, expired, or malformed tokens were rejected with a 401 Unauthorized response.

2. Security Controls Applied

- **HTTPS Only:**
  - The Cloudflare Gateway automatically enforced HTTPS connections.
- **Rate Limiting:**
  - Attempted using a Cloudflare Durable Object to track request counts per IP.
  - Limited to 10 requests per 10 seconds per client.
  - Due to Cloudflare constraints on free accounts, rate limiting was noted but could not be fully enforced in production testing.
- **Access Control:**
  - Roles/scopes encoded within JWT payload

- Backend API validated the JWT payload to ensure only authorized users could access sensitive endpoints.
- Requests missing the required scope were rejected with a 403 Forbidden response.

Setup and Workflow

1. **API Gateway Creation:**
   - Cloudflare Worker deployed as the gateway (worker.js)
   - Gateway routes incoming requests to the backend API
2. **JWT Validation:**
   - Gateway fetches JWKS from the identity provider
   - Uses crypto.subtle to verify JWT signatures
   - Checks claims (iss, aud, exp)
3. **Backend API Connection:**
   - API Gateway forwards validated requests to the posts API endpoint
   - Backend responses returned to the client
4. **Security Enforcement:**
   - HTTPS enforced by Cloudflare
   - Access control via JWT scopes
   - Rate limiting (experimental, Cloudflare Durable Object)

Observed Behavior

Successful Requests:

- Requests with valid JWTs returned the expected posts JSON response.

Unauthorized Requests:

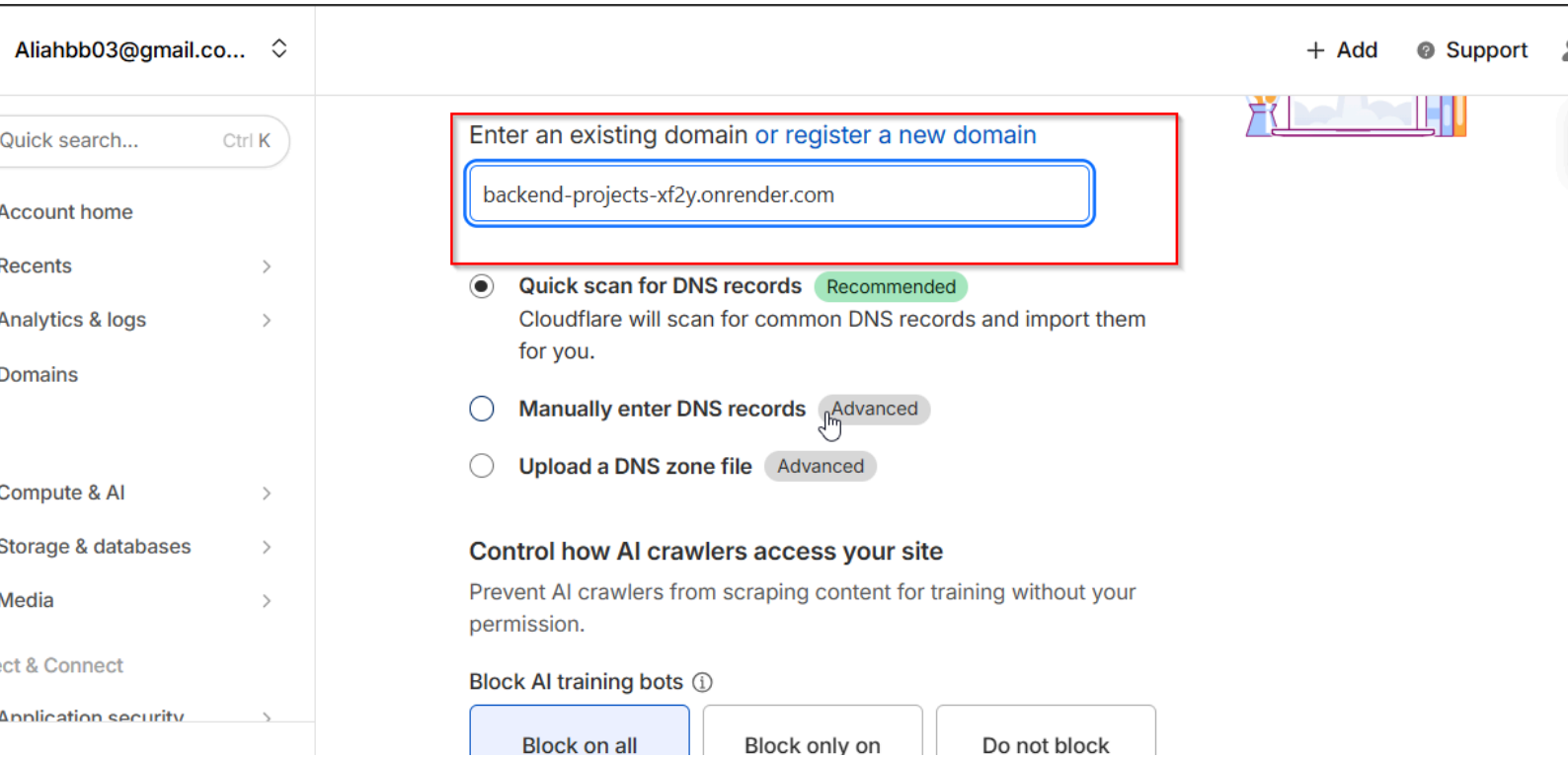
- Requests with no JWT, expired tokens, or invalid signatures returned 401 Unauthorized.
- Requests with missing roles/scopes returned 403 Forbidden.

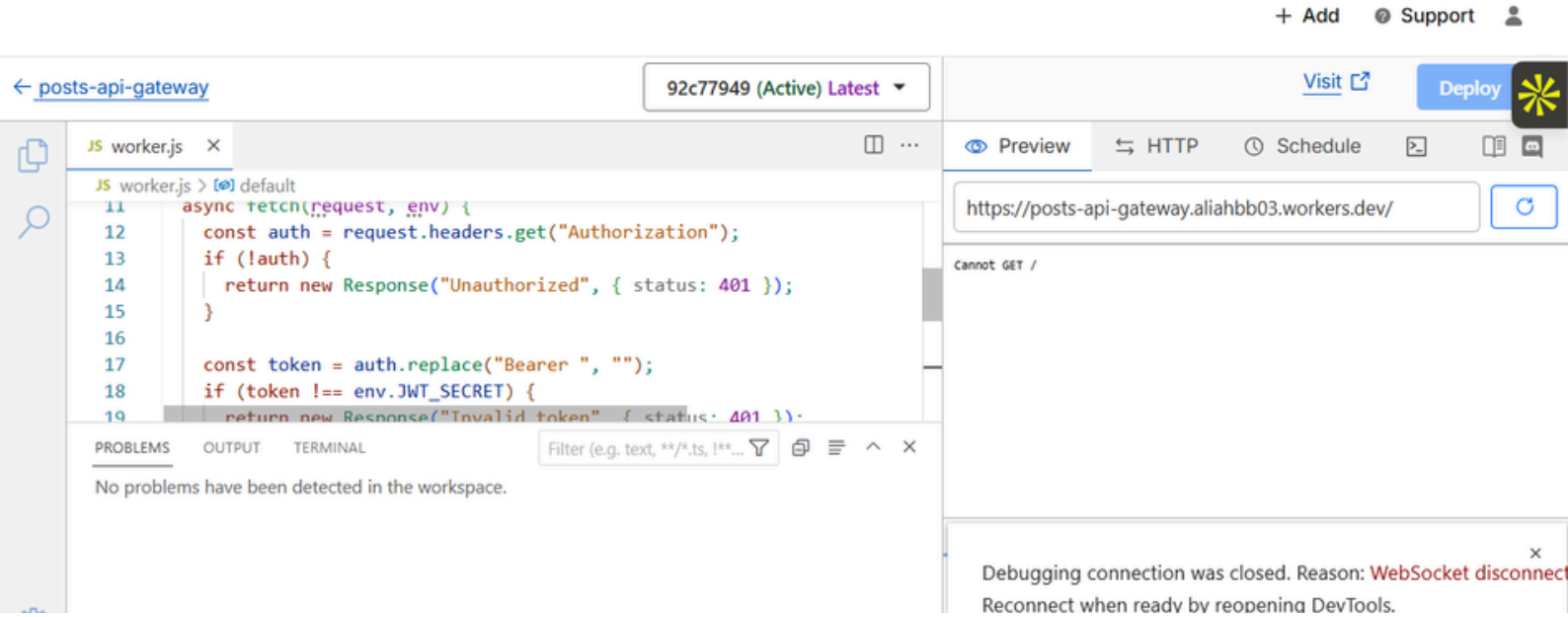
Rate Limiting (Tested Locally):

- Attempting more than 10 requests within 10 seconds triggered a 429 Too Many Requests response in local testing.

Proof of concept

1. Cloudflare Worker Gateway Deployment:





(Screenshot showing the worker deployed in Cloudflare dashboard)

2. Postman Request with Valid JWT:

DECODED HEADER

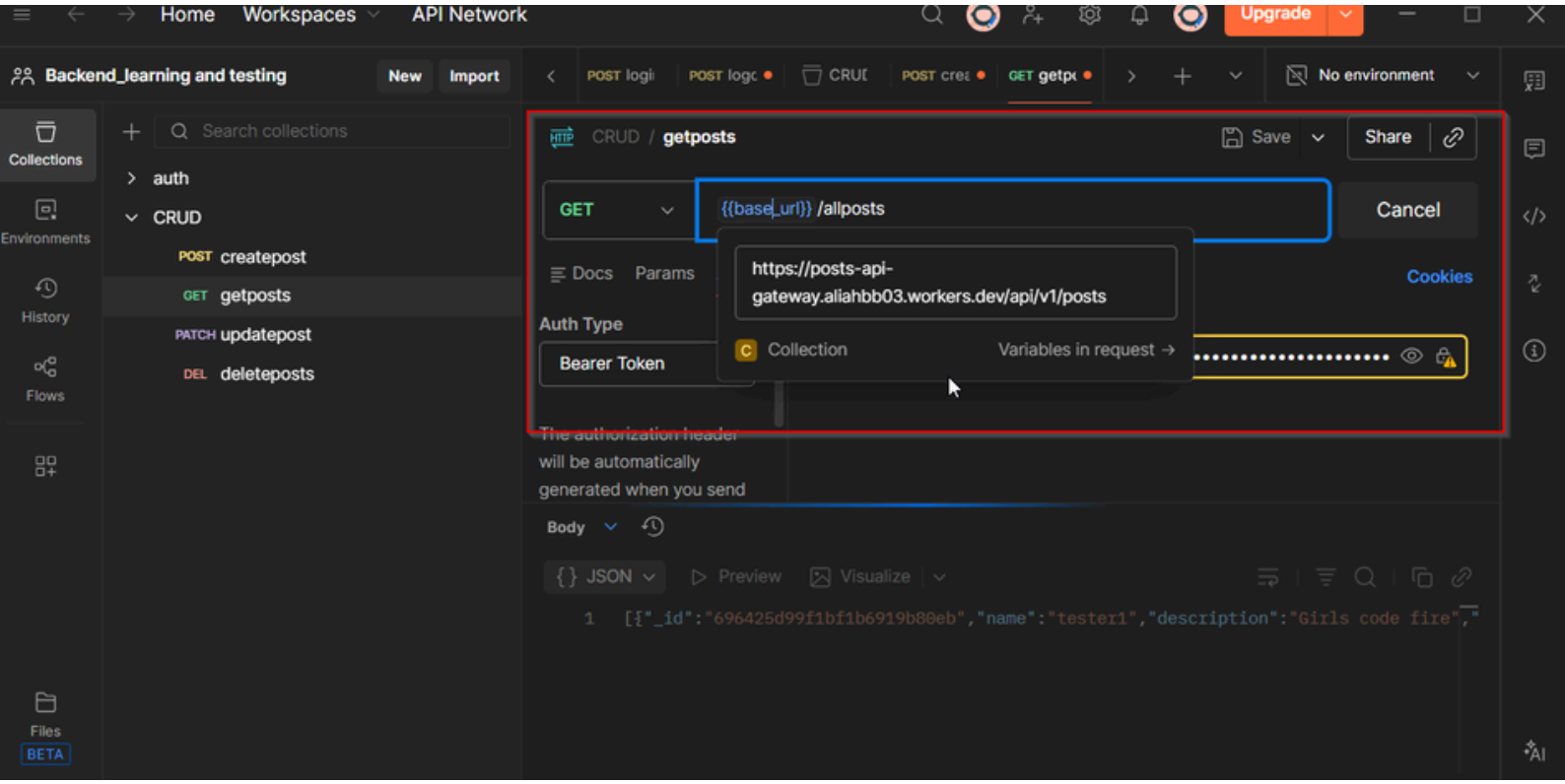
JSON CLAIMS TABLE COPY

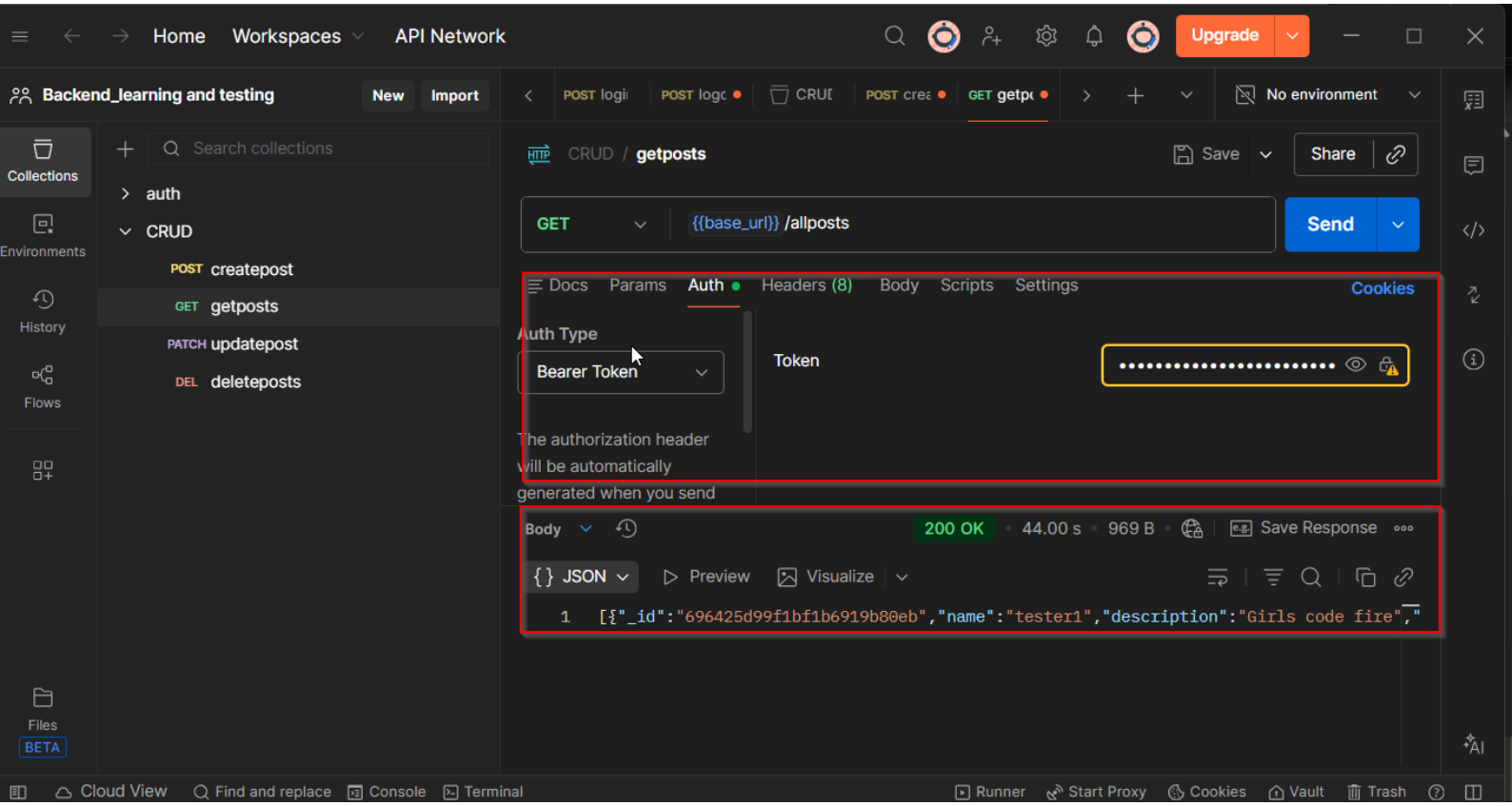

```
{
  "alg": "RS256",
  "typ": "JWT",
  "kid": "kkuElR4hbP4XerrduBtYo"
}
```


DECODED PAYLOAD

JSON CLAIMS TABLE COPY

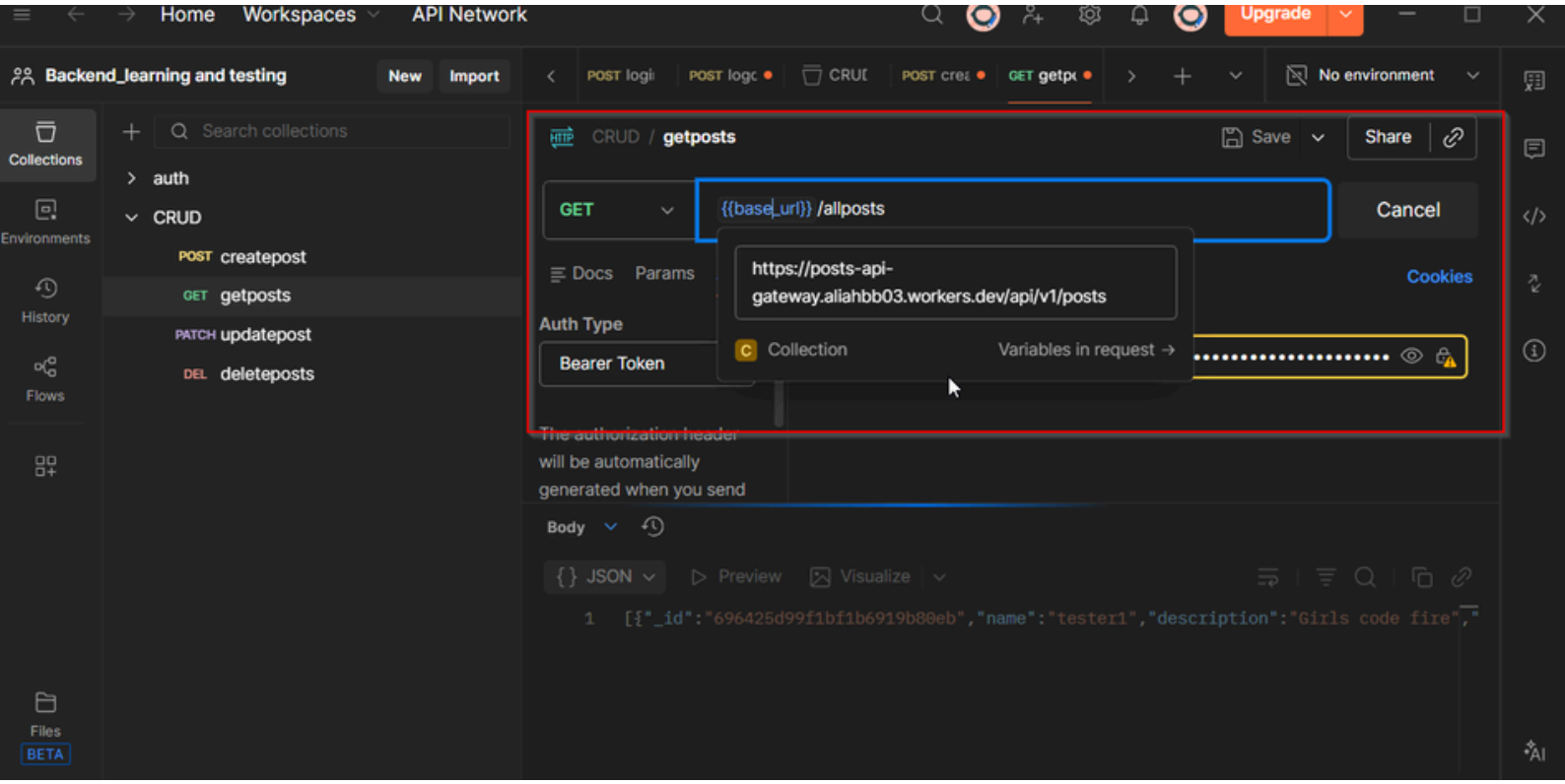

```
{
  "iss": "https://dev-yaoca5kyfa7tnya4.us.auth0.com/",
  "sub": "LHIWbKBq9bUat7M91tWaRUpZaCSKaNXN@clients",
  "aud": "https://posts-api-gateway.aliahbb03.workers.dev",
  "iat": 1768738006,
  "exp": 1768824406,
  "gty": "client-credentials",
  "azp": "LHIWbKBq9bUat7M91tWaRUpZaCSKaNXN"
}
```

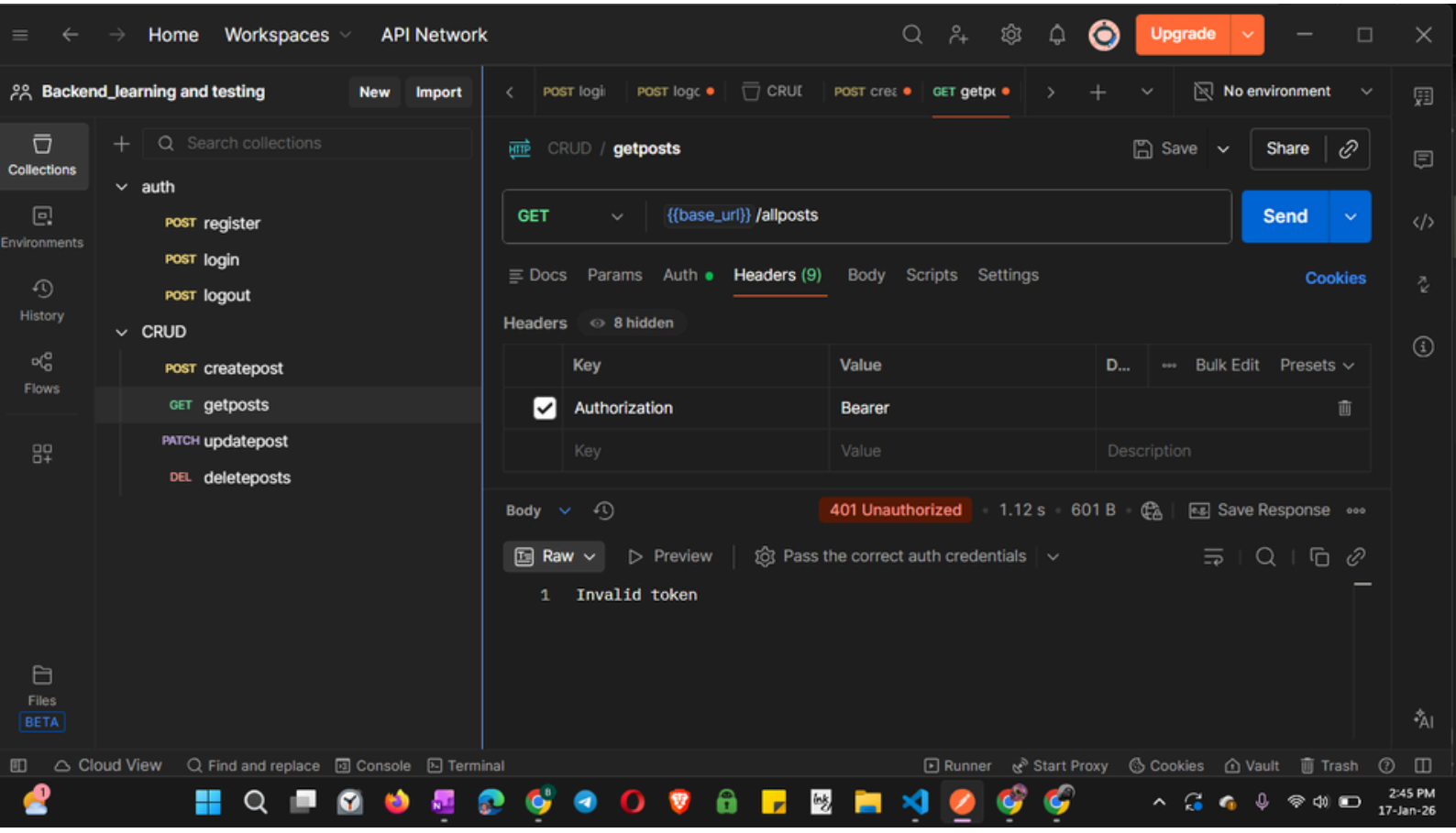




(Screenshot showing 200 OK response and JSON data returned from the backend)

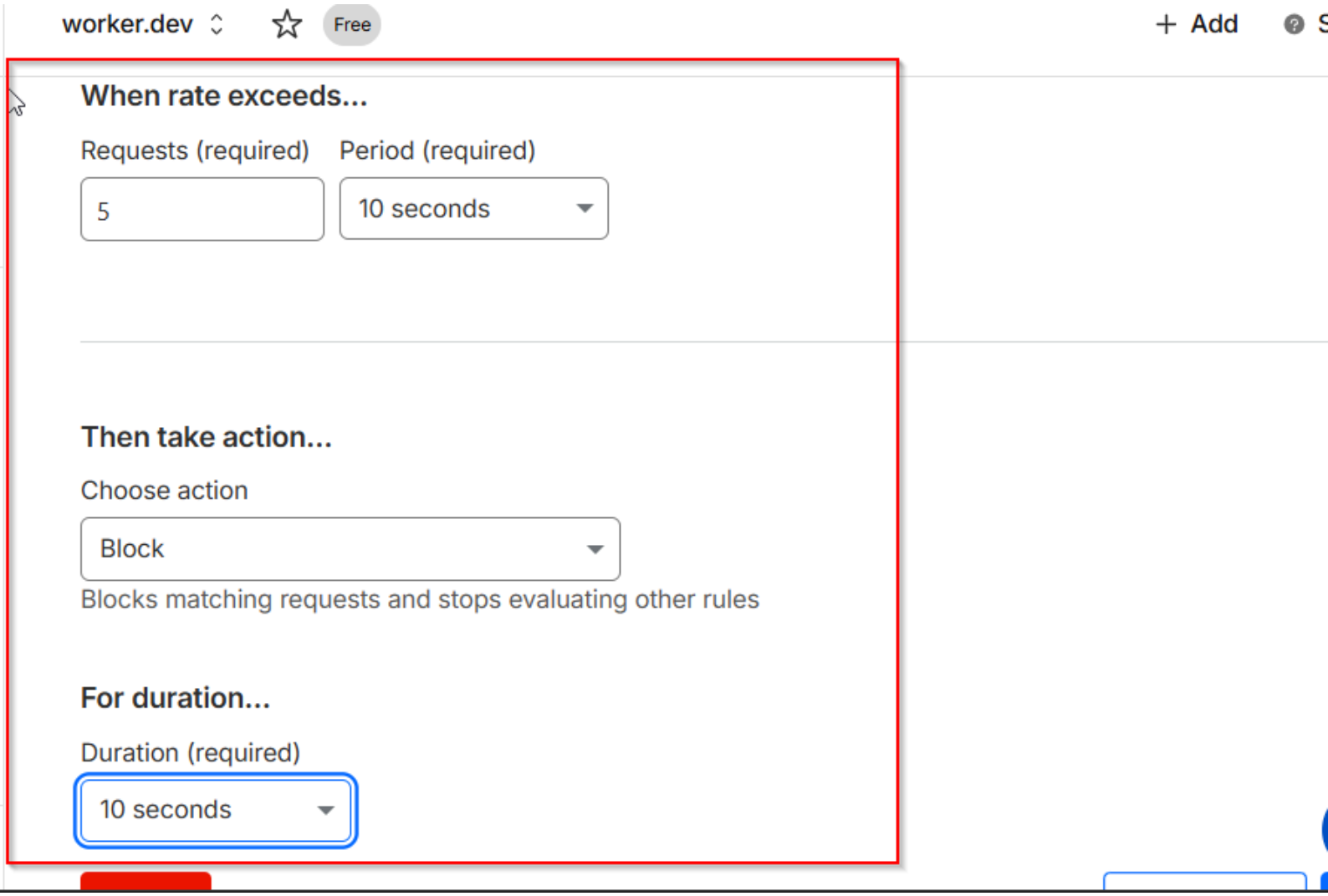
3. Postman Request with Invalid JWT:

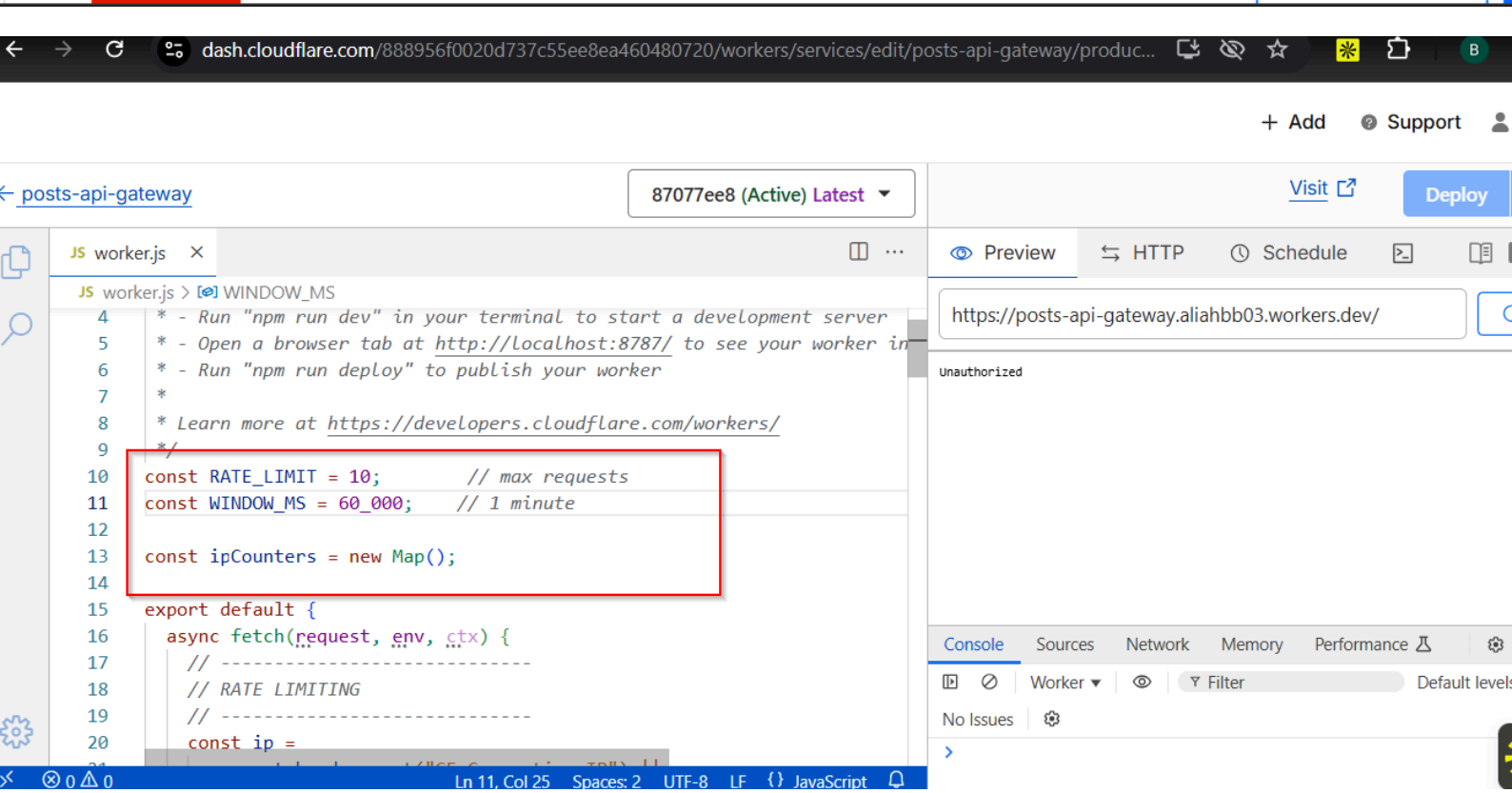




(Screenshot showing 401 Unauthorized response)

4. Attempted Local Rate Limiting Test:





Nothing seems to work

(Screenshot of 429 Too Many Requests triggered after exceeding 10 requests in 10 seconds, that did not work out)

Challenges and Notes

- Cloudflare free tier does not fully enforce Durable Object-based rate limiting across distributed workers.
- Authentication and audience checks were fully functional, providing secure access to backend endpoints.
- HTTPS enforcement ensured encrypted traffic between clients and the API Gateway.

Conclusion

The lab successfully demonstrated:

- Deployment of a cloud-based API Gateway using Cloudflare
- JWT-based authentication with issuer, audience, and expiration validation
- Access control enforced via JWT audience
- Basic security measures including HTTPS and experimental rate limiting

Overall Security Posture: Secure with minor limitations on distributed rate limiting due to platform constraints.